

SAFE WHERE INC.



INNOVATION BUSINESS PLAN

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I. Executive Summary



SAFE WHERE
Show your care with Safe Where

1

Innovation Overview:

Safe Where offers **peace of mind** in a **bracelet**. This **child-friendly, waterproof** accessory combines style with cutting-edge **satellite tracking technology**, allowing parents to discreetly monitor their child's location. Its unique design features fun, kid-themed beads, and **animal tracking**, making it appealing for kids to wear. With Safe Where, we're tackling the critical issue of child abductions in the U.S., offering a smart, covert way to keep your children safe.

Problems:

I. Preventable Kidnappings: Imagine that every 40 seconds, a child goes missing in the U.S. It's a parent's worst nightmare. Current Amber Alert systems yielded a 13% effective in these scary situations, leaving families heartbroken.

II. Impracticality of Existing Jewelry Trackers:

A. Not Watertight: Popular trackers like the Apple Air Tag and Tile bracelets are bulky and not fully waterproof. One little splash, and they might stop working!

B. Low Quality, High Risk: Cheaper options are out there, but they're made with harmful materials. Some have phthalates; hormone-disrupting chemicals that pose a risk of early-onset cancer.

C. Boring for Kids: Most trackers are just... trackers. They're not fun, therefore kids don't want to wear them. NCBI discovered a 20% increase in children who wanted to stop wearing GPS-tracking bracelets.

D. Need a Phone: 47% of kids can't even be tracked because they need a phone for these gadgets to work.

E. No Signal, No Safety: In places without cell service, current trackers capitulate. They can't send alerts or reach out to help when kids need it most.

Solution:

Safe Where uses **high-quality** beaded bracelets that include **child and animal tracking**.

Embedded with a Nano Spider GPS tracker and a miniature panic button, tracking is hidden from the kidnapper's eye. In emergencies, your child can press the **panic button**, sending an alert notification to emergency contacts. Using a **synced GPS app**, parents can track their child's location and location history any time.

Unique Value Proposition:

I. Fashionable and Discreet: Unlike bulky and noticable tracking bracelets, Safe Where is stylish and includes a customizable beaded beacelet that kids want to wear.

II. Innovative Tracking Technology: It's packed with advanced 24/7 child tracking and a fun animal tracking feature that keeps kids engaged!

III. Top-Notch Security Features: Our bracelet comes with satellite GPS tracking, a secure barrel clasp that keeps the bracelet firmly in place, and a rechargeable battery, so you never lose your child's GPS location.

Customer Segment:

I. Primary Market– Parents with young children ages 5–11:

The ideal customer in this segment resides in the United States and is a suburban family with an income over \$55,000 concerned with their child's safety.

II. Secondary Market– Elementary and Middle Schools:

Elementary and Middle Schools lacking safety for children. With over 98,000 public schools in the U.S., even minimal adoption rates in this sector can significantly increase our market reach and revenue.

III. Market Size: In the United States alone, the child safety and wearable technology market is currently valued at over \$1 billion, with an expected annual growth rate of 8%. This growth is driven by increasing concerns about child safety among parents.

Finances:

Safe Where operates on a **model of product sales** with a subscription service for ongoing updates and support. We scale and grow through strategic marketing and partnerships, focusing on expanding our market reach and continuously improving our technology.

The business requires a total investment of **\$574,618**. A **loan of \$150,000** will be needed and will be repaid in 2 years with a **6.5% interest rate**.



II. PROBLEM

How Do I Keep My Child Safe?



The pressing issue we aim to tackle is the rising incidents of children getting lost, followed, or abducted, causing immense distress among parents who fear for their children's safety. Recognizing this widespread problem, Safe Where embarked on a mission to create innovative and interactive bracelets that not only ensure the security of children but also alleviate the stress parents experience when their child is missing. Our motivation stems from our desire to provide a safer environment for the future generation of children.



AMBER Alert
Child abduction in
Chevrolet Tahoe T
KVZ-1194. Victim
Hispanic/Male, lar
seen wearing unk
Suspect is 38 -yea
Juan Aguilar-Cand
information.

A significant challenge we faced was convincing children to wear our safety bracelet. Through collaborative brainstorming sessions with our mentors, we constructed a unique approach. Safe Where has implemented tracking endangered species into our solution. Now, children can not only be tracked by their parents but can also engage with the bracelet by tracking the journey their new animal best friend. This innovative feature establishes a meaningful connection between the child and the endangered species, fostering a sense of bond and responsibility.

While our product primarily addresses the aforementioned issues, our research has unveiled additional benefits. In addition to enhancing child safety, our solution spreads awareness about endangered species around the world and supports their conservation efforts. Furthermore, to combat the issues we are tackling, the Safe Where team donates part of our proceeds to The Polaris Project and an anti-human trafficking organization, making our company more comprehensive and impactful.

http://



www.safe-where.com

II. Problem

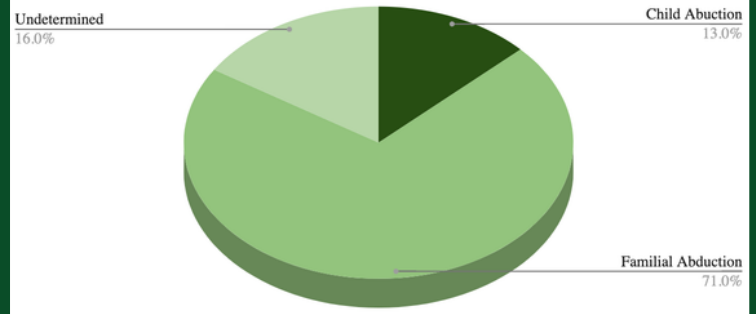


Preventable Kidnappings:

“Every forty seconds, a child goes missing” in the United States. Over 2,000 children are reported missing every day, totaling more than 800,000 missing children annually, as per the F.B.I. These devastating situations, where children’s lives are at risk, occur due to the lack of a secure and discreet method for parents to track their children. The delayed response of Amber Alerts keep an immediate response from happening.

The National Center for Missing and Exploited Children reported a 71% effectiveness rate for Amber Alerts in familial cases as of 2021. However, in instances involving abduction by strangers, Amber Alerts only yield a 13% success rate in recovering a child within three hours of the abduction (Griffin, 2021). In essence, Amber alerts are significantly less effective in cases of child abduction by strangers. When children face these dangerous situations, it is imperative to have effective solutions in place to safeguard and rescue them, offer parents peace of mind, and reduce the likelihood of children falling into perilous situations.

Amber Alert Effectiveness Rate



Impracticality of Existing Jewelry Trackers:

Unreliable, Unsecure, and Low-cost Tracking:

Existing location tracking bracelets such as the Apple Air Tag Bracelets and Tile GPS Tracking Bracelets pose many problems for effective tracking. With the feature of interchangeable GPS trackers, the trackers are not secured tightly, and water seeps into the tracker and band. Because the trackers can be removed with ease, such devices are not durable enough to be suitable for a child to wear. Moreover, priced between \$12.00 and \$20.00, existing trackers like the Air Tag Bracelets are not created with high-quality materials. With materials such as silicon used for the band of the bracelet, if manufactured cheaply, could be linked to an increased incidence of certain cancer types. Researchers at Duke University tested different silicon wristbands and found that at least all bands tested had traces of phthalates, which are “hormone-disrupting chemicals.” Thus, existing bracelets could pose a risk of early-onset cancer (Safety And Health Magazine, 2022).

Young Children Need Phones to be Tracked:

Current location tracking bracelets with non-removable trackers and other band types are also not kid-friendly due to the huge size of the tracker, uncomfortableness, and lack of discreetness. All existing trackers also need a phone to function properly. As young children are the ones who are using such devices, and only 53% of children up to age 11 own a phone, it becomes infeasible for children without phones to be tracked, accounting for 47% of children (NPR.org, 2019). This leaves the market with a high demand for discreet tracking devices without the child needing a phone.

Lack of Incentive:

Furthermore, tracking jewelry such as Apple Air Tag Bracelets come in watch stylings and only have the tracking feature. This presents a problem for the child to continue wearing the product because there is no incentive for them to wear it– no telling time, seeing messages, or tracking an animal friend. In fact, The National Center for Biotechnology Information conducted a 4 week study and found that as the weeks progressed, there was a 20% increase in children who wanted to stop wearing it from loss of interest, “too bulky and restrictive from sports,” or “having lost it” (NCBI, 2019).



Children Are Not Being Protected When Out Of Cell Tower Range:

When children are in a dangerous situation and there is no cellular data present, the most help existing jewelry trackers can do is send an automatic alert to parents, letting them know that the alert wasn’t able to be sent. For instance, take Invisawear, a jewelry tracking company targeted toward parents. They pair their jewelry to their app via Bluetooth tracking and use cellular data to send their emergency alerts. While this is helpful when cellular data is present, if out of range, children are not able to receive the help they need by contacting authorities, thus leaving a gap in the market for satellite tracking products.



III. Customer Segment

TARGET MARKET

Our product is accessible to customers of all age groups. However, in our first year, our focus lies on concerned parents who have children ages 5-11 within the United States. In shaping our target market, we considered the market size and the demographics of younger children who do not have access to electronic devices. This thorough analysis and in-depth market research have enabled us to define a precise and achievable target audience (1.1 Appendix).

Safe Where bracelets will be strategically promoted across social media platforms, including Twitter, Tiktok, Instagram, and joined Facebook groups to enhance our product's visibility and outreach. By employing an effective search engine marketing strategy, consumers can effortlessly locate our product by typing in keywords such as 'tracking jewelry' or 'discreet bracelets' and receive seamless access to the Safe Where bracelets for potential consumers and users. These marketing tactics are intended to foster widespread familiarity with the unique benefits that Safe Where bracelets offers.

WHERE DO THE NUMBERS COME FROM?

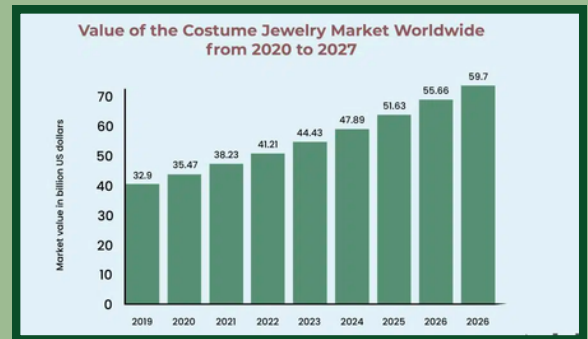
Number of Kids in the U.S. (5-11): 28,102,306

Number of Total Parents in the U.S.: 63.1 million recorded in 2020

Percent of Kids Without a Smart Device (younger than 11): 83%

Jewelry Market Size Value: 38.23 billion in the year of 2021

Market Size Growth: Surpass 75.73 billion by 2030



WHY THE UNITED STATES?

We have chosen to release Safe Where in the United States geographic market due to it being the largest geographic market in the world. The United States is in forefront of opening world markets, thereby helping us protect more children at risk of harm. In our second year, Safe Where will broaden the market to the following international market's that are plagued by child abductions: South Asia, Sub-Saharan Africa, Canada, and Europe, will triple our total addressable market. After global expansion, we may consider expanding Safe Where to teenagers ages 12-15.

Primary Market

Demographic: Adults to child (25+)

Annual Income: \$55,000+

Geographic: Suburban area

Secondary Market

Safe Where products will be sold in various stores, on websites, and to jewelers such as Crowned Free and Fahlo (our partnerships). To further fulfill marketing tactics, we will engage in activities with Gen Alpha by attending primary and secondary school events and marketing Safe Where bracelets through the school newsletters and various assemblies.

IV. Unique Value Proposition

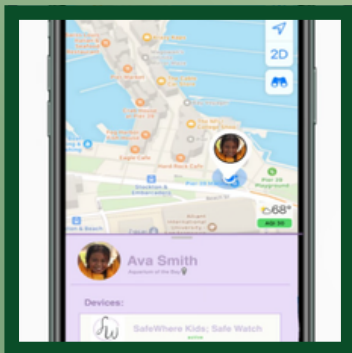


Product features

Unlike any of our competitors, Safe Where provides the child with a discreet way of calling for help when in a dangerous situation. While other tracking companies' devices are too bulky and noticeable, our tracking device is embedded safely inside a bracelet bead, which cannot be removed or visible. Not only do Safe Where bracelets provide the tracking feature, but to fully ensure the prevention of kidnapping, we have embedded a panic button feature on the back of the bracelet bead. If the child is aware that they are in a dangerous situation, they can press their panic button twice to contact emergency services and their three chosen emergency contacts on the Safe Where app.



iPhone App



Our app is another essential part of our product. By downloading the my Safe Where app on the phone's app store, parents are able to connect their child's bracelet to the app and view their child's bracelet location on a satellite map at all times. While setting up the app, parents can choose up to three emergency contacts they would like to store in case of an emergency. Whereas Life360's app is only for iPhone users, the mySafeWhere app is available to not only iPhone users but Samsung and Google Pixel users as well.

Animal tracking

A child might not be motivated to wear such a bracelet, which is why we have included the kid-friendly feature of tracking an animal best friend. While ordering the bracelet, the parent and child are allowed to select an endangered animal of their choice to track on the FAHLO website. FAHLO is a company specializing in animal tracking that we are partnering with to include this feature with the bracelet. This animal will also be featured as a charm on the bracelet so that while the parent is able to track their child, the child can track their new animal best friend online.



GPS tracker chip

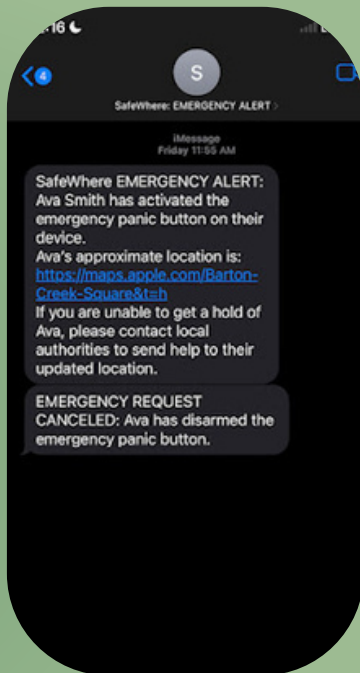
Our bracelet is implemented with the smallest GPS chip on the market: the Nano-spider (1.2 Appendix). It serves as a typical tracker, connecting to a series of satellites to determine the child's location. The Nano-spider chip is 7x7 mm, which fits inside the animal charm and is perfect for discreetness (1.3 Appendix). The GPS is made by OriginGPS, an IoT company that creates cellular and satellite products.

Rechargeable battery

Each bracelet also comes with a miniature lithium ion battery embedded in the animal charm, which is responsible for the charging of the bracelet. The battery life lasts between 4-5 days, providing a longer wear time compared to existing trackers on the market. On the bottom of the animal charm, there is a small charging port where the portable USB charger fits into. This intricately crafted USB charger is included with the purchase of the bracelet.

Discreet, Effective, Customizable:

Safe Where was developed to address and decrease global kidnapping rates. Each bracelet contains the world's tiniest fully integrated IOT GPS tracker, eliminating concerns about its bulkiness. Our product offers reliable tracking technology while also specializing in discreetness. Crafted with high-quality, customizable beading, each bracelet is made to the user's preferences, encouraging regular wear.



Automated Emergency Notification:

Upon double-pressing the panic button, the designated emergency contacts – up to three – will receive an automated message containing the child's name, coordinates, and additional information about the bracelet's activation. In instances where the panic button is accidentally armed, holding it down will cancel the request. After disarming the bracelet, parents will be prompted with an auto-generated text confirming the cancellation of the panic button request.

No Service, No Problem:

Each bracelet will come equipped with a state-of-the-art GPS tracker measuring 7mm x 7mm in size. The tracker, identified as ORG4572, belongs to the OriginGPS spider series and harnesses the power of satellite tracking technology to provide highly accurate and reliable IP location results. Moreover, the bracelet will continue to function

seamlessly even with no cellular service. As the GPS information is gathered through satellites located all around the globe, information will be stored and sent directly to the appropriate phone.

Safe Where in Action:

Safe Where's website will be the direct platform for purchasing bracelets. Following purchase, users will be guided to the Safe Where app to set up the bracelets. This process involves creating an account with a first and last name, email, phone number, selecting a subscription plan, specifying the number of bracelets that will be tracked, the name of the user being tracked, pairing the bracelet and phone with a custom code received upon bracelet arrival, selecting up to three emergency contacts, and identifying up to 25 frequently visited locations such as homes, schools, places of worship, and activities. Once setup is complete, parents can effortlessly navigate the app and accurately track their child's location.

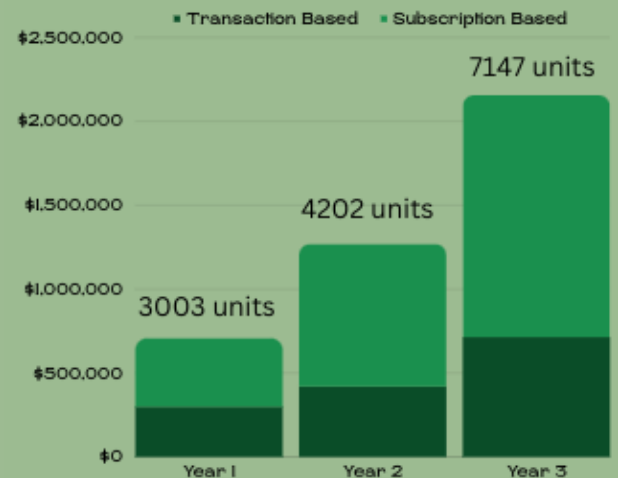


VI. Conclusion



Safe Where is something that has never been brought to the market before, and truly is a unique and revolutionary product that will save the lives of children in a discreet and fashionable manner. Safe Where bracelets cover all of the following issues:

- **Child Doesn't Have a Phone for Discreet Location Tracking?** Parents see their child's location through the discreetness of the embedded GPS chip inside the bracelet bead. The bracelet connects to the parent's phone.
- **No Way to Contact Home?** When double-clicked, the panic button feature immediately contacts emergency contacts and emergency services through the app.
- **Child Doesn't Want to Wear It?** We have made the bracelet fun for the child to wear with the incentive of tracking their animal best friend through our FAHLO partnership.
- **Other Jewelry too Fragile?** Safe Where designed the bracelets to have a screw barrel clasp design for durability and security on the child's wrist.
- **Other Devices not Kid-friendly?** Safe Where has a multitude of color and charm options for the child to choose from, which are fashionable for all kids.



Revenue Projections

The total company investment of \$574,618 will be spent on the approval of patents, the development of our viable product and GPS app, as well as development of a strong brand image through marketing platforms, venues, and search engine marketing. The stage one startup cost of \$38,427 includes \$2,000 of angel investments. The \$534,191 remaining amount will be generated through the company's first three years of operations and borrowed from banks. We would like to request a \$150,000 loan of 6.5% interest over a span of 2 years starting January 2024. This specific loan will be directed toward research and development and marketing expenses.

Safe Where will help combat the issue and reduce the amount of child abductions all over the world. We will market this innovative product through our jewelry company partnerships like FAHLO and Crowned Free, social media platforms to reach parents, and elementary-middle schools to educate the children. Safe Where is in the developmental stages of taking our product from prototype to market, and we are on the way to launching our bracelets to families all over the world. By working with partnership companies and marketing to parents, we will ensure the safety of the future generation of children from any dangerous situations. We aim to see every parent reassured of their child's security and see every child wear a Safe Where bracelet on their wrist.

VII. Bibliography

- >> Auxier, Brooke. "Children's engagement with digital devices, screen time." Pew Research Center, 28 July 2020. <https://www.pewresearch.org/internet/2020/07/28/childrens-engagement-with-digital-devices-screen-time/>.
- >> Annie E. "Child population by age group | KIDS COUNT Data Center." CHILD POPULATION BY AGE GROUP IN UNITED STATES, July 2022, <https://datacenter.aecf.org/data/tables/101-child-population-by-age-group#detailed/1/any/false/1095,2048,574.1729,37.871,870,573,869,36/62.63,64.6,4693/419,420>.
- >> Child Watch of North America. "Statistics." Child Watch of North America, May 2021, <https://childwatch.org/statistics/>.
- >> Edney, Sarah, et al. "Parental Perspectives of a Wearable Activity Tracker for Children Younger Than 13 Years: Acceptability and Usability Study." NCBI, 4 November 2019, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6861996/>.
- >> Facts & Factors. "[Latest] Global Jewelry Market Size/Share Worth USD 517.27 Billion by 2030 at a 8.25% CAGR: Facts & Factors (Industrial Trends, Report, Analysis, Players, Outlook, Statistics, Forecast, Growth, Value, Segmentation)." GlobeNewswire, 13 February 2023, <https://www.globenewswire.com/en/news-release/2023/02/13/2606713/0/en/Latest-Global-Jewelry-Market-Size-Share-Worth-USD-517-27-Billion-by-2030-at-a-8-25-CAGR-Facts-Factors-Industrial-Trends-Report-Analysis-Players-Outlook-Statistics-Forecast-Growth-V.html#>.
- >> Friedrich, Mike. "New Estimates on America's Families and Living Arrangements." U.S. Census Bureau, 2 December 2020, <https://www.census.gov/newsroom/press-releases/2020/estimates-families-living-arrangements.html>.
- >> Griffin, T., Williams, J. H., & Kadleck, C. (2022). AMBER Alert Effectiveness Reexamined. Criminal Justice Policy Review, 33(1), 23-44. <https://doi.org/10.1177/08874034211026366>
- >> Halpern, Mollie. "FBI and the National Center for Missing & Exploited Children — FBI." FBI, 12 July 2011, https://www.fbi.gov/audio-repository/news-podcasts-inside-inside_071211.mp3/view.
- >> Kamenetz, Anya. "New Survey Examines Phone, Media Use Among U.S. Teens And Children." NPR, 31 October 2019, <https://www.npr.org/2019/10/31/77483891/its-a-smartphone-life-more-than-half-of-u-s-children-now-have-one>.
- >> Kraut, Michael E. "Missing and Abducted Children." Child Crime Prevention & Safety Center, November 2022, <https://childsafety.losangelescriminallawyer.pro/missing-and-abducted-children.html>.
- >> Safety And Health Magazine. "Firefighter study shows inexpensive silicone wristbands can help track chemical exposures." Safety+Health Magazine, 11 July 2022, <https://www.safetyandhealthmagazine.com/articles/22776-firefighter-study-shows-inexpensive-silicone-wristbands-can-help-track-chemical-exposures>.
- >> Tai, Katherine. "Economy & Trade | United States Trade Representative." USTR, January 2023, <https://ustr.gov/issue-areas/economy-trade>.
- >> World Populations Review. "Kidnappings per Country 2023." World Population Review: Kidnappings per Country, March 2023, <https://worldpopulationsreview.com/country-rankings/kidnappings-per-country>.

VIII. Appendix

1.1 Interview Responses

Interviewee	Topic	Notes	Summary	Key Points	Follow-up
1	Interview 1: 11-11-2020	Topic: Interview 1: 11-11-2020	Notes: Interview 1: 11-11-2020	Key Points: Interview 1: 11-11-2020	Follow-up: Interview 1: 11-11-2020
2	Interview 2: 11-11-2020	Topic: Interview 2: 11-11-2020	Notes: Interview 2: 11-11-2020	Key Points: Interview 2: 11-11-2020	Follow-up: Interview 2: 11-11-2020
3	Interview 3: 11-11-2020	Topic: Interview 3: 11-11-2020	Notes: Interview 3: 11-11-2020	Key Points: Interview 3: 11-11-2020	Follow-up: Interview 3: 11-11-2020
4	Interview 4: 11-11-2020	Topic: Interview 4: 11-11-2020	Notes: Interview 4: 11-11-2020	Key Points: Interview 4: 11-11-2020	Follow-up: Interview 4: 11-11-2020
5	Interview 5: 11-11-2020	Topic: Interview 5: 11-11-2020	Notes: Interview 5: 11-11-2020	Key Points: Interview 5: 11-11-2020	Follow-up: Interview 5: 11-11-2020
6	Interview 6: 11-11-2020	Topic: Interview 6: 11-11-2020	Notes: Interview 6: 11-11-2020	Key Points: Interview 6: 11-11-2020	Follow-up: Interview 6: 11-11-2020
7	Interview 7: 11-11-2020	Topic: Interview 7: 11-11-2020	Notes: Interview 7: 11-11-2020	Key Points: Interview 7: 11-11-2020	Follow-up: Interview 7: 11-11-2020
8	Interview 8: 11-11-2020	Topic: Interview 8: 11-11-2020	Notes: Interview 8: 11-11-2020	Key Points: Interview 8: 11-11-2020	Follow-up: Interview 8: 11-11-2020
9	Interview 9: 11-11-2020	Topic: Interview 9: 11-11-2020	Notes: Interview 9: 11-11-2020	Key Points: Interview 9: 11-11-2020	Follow-up: Interview 9: 11-11-2020
10	Interview 10: 11-11-2020	Topic: Interview 10: 11-11-2020	Notes: Interview 10: 11-11-2020	Key Points: Interview 10: 11-11-2020	Follow-up: Interview 10: 11-11-2020
11	Interview 11: 11-11-2020	Topic: Interview 11: 11-11-2020	Notes: Interview 11: 11-11-2020	Key Points: Interview 11: 11-11-2020	Follow-up: Interview 11: 11-11-2020
12	Interview 12: 11-11-2020	Topic: Interview 12: 11-11-2020	Notes: Interview 12: 11-11-2020	Key Points: Interview 12: 11-11-2020	Follow-up: Interview 12: 11-11-2020
13	Interview 13: 11-11-2020	Topic: Interview 13: 11-11-2020	Notes: Interview 13: 11-11-2020	Key Points: Interview 13: 11-11-2020	Follow-up: Interview 13: 11-11-2020
14	Interview 14: 11-11-2020	Topic: Interview 14: 11-11-2020	Notes: Interview 14: 11-11-2020	Key Points: Interview 14: 11-11-2020	Follow-up: Interview 14: 11-11-2020
15	Interview 15: 11-11-2020	Topic: Interview 15: 11-11-2020	Notes: Interview 15: 11-11-2020	Key Points: Interview 15: 11-11-2020	Follow-up: Interview 15: 11-11-2020
16	Interview 16: 11-11-2020	Topic: Interview 16: 11-11-2020	Notes: Interview 16: 11-11-2020	Key Points: Interview 16: 11-11-2020	Follow-up: Interview 16: 11-11-2020
17	Interview 17: 11-11-2020	Topic: Interview 17: 11-11-2020	Notes: Interview 17: 11-11-2020	Key Points: Interview 17: 11-11-2020	Follow-up: Interview 17: 11-11-2020
18	Interview 18: 11-11-2020	Topic: Interview 18: 11-11-2020	Notes: Interview 18: 11-11-2020	Key Points: Interview 18: 11-11-2020	Follow-up: Interview 18: 11-11-2020
19	Interview 19: 11-11-2020	Topic: Interview 19: 11-11-2020	Notes: Interview 19: 11-11-2020	Key Points: Interview 19: 11-11-2020	Follow-up: Interview 19: 11-11-2020
20	Interview 20: 11-11-2020	Topic: Interview 20: 11-11-2020	Notes: Interview 20: 11-11-2020	Key Points: Interview 20: 11-11-2020	Follow-up: Interview 20: 11-11-2020

1.2 Nano-spider Information

OriginGPS
NanoSpider

TL Transmission-Transistor Logic
UART Universal Asynchronous Receiver/Transmitter
VCI Voluntary Control Council for Interference by Information Technology equipment
VEP Vertical Error Probability
VGA Variable Gain Amplifier
WAAS Wide Area Augmentation System

9. ABOUT SPIDER FAMILY
OriginGPS GNSS receiver modules have been designed to address markets where size, weight, stand alone operation, highest level of integration, power consumption and design flexibility - all are very important. OriginGPS' Spider family breaks size barrier, offering the industry's smallest fully-integrated, highly-sensitive GPS and GNSS modules.
Spider family features OriginGPS' proprietary N2™ technology for high sensitivity and noise immunity even under marginal signal condition, commonly found in urban canyons, under dense foliage or when the receiver's position in space rapidly changes.
Spider family enables the shortest TTFF (Time-To-First-Fix) with minimal design risks.
Just connect an antenna and power supply on a 2-layer PCB.

10. ABOUT NANO SPIDER MODULE
Nano Spider is a complete SIP featuring LGA SMT footprint designed to commit unique integration features for high volume cost sensitive applications.
Designed to support ultra-compact applications such as smart watches, wearable devices, trackers and digital cameras, Nano Spider ORGA400 module is a miniature multi-channel GPS with SBAS, QZSS and other regional overlay systems receiver that continuously tracks all satellites in view, providing real-time positioning data in industry's standard NMEA format.
Nano Spider ORGA400 module offers superior sensitivity and outstanding performance, achieving rapid TTFF in less than one second, accuracy of approximately two-meters, and tracking sensitivity of -165dBm.
Sized only 4.1mm x 4.1mm Nano Spider ORGA400 module is industry's small sized, record breaking solution. ORGA400 module integrates LNA, SAW filter, TCXO, RTC crystal shield with market leading SiRFstar™ GPS SoC.
Nano Spider ORGA400 module is introducing industry's lowest energy per fix ratio, unparalleled accuracy and extremely fast time to first fix even under challenging signal conditions, such as in built-up urban areas, dense foliage or even indoors.
Integrated GPS SoC incorporating high performance microprocessor and sophisticated firmware keeps positioning payload off the host, allowing integration in embedded solutions with low computing resources. Innovative architecture can detect changes in context, temperature, and satellite signals to achieve a state of near continuous availability by maintaining and opportunistically updating its internal fine time, frequency, and satellite ephemeris data while consuming mere microwatts of battery power.

11. ABOUT ORIGINGPS
OriginGPS is a world leading designer, manufacturer and supplier of miniature positioning modules, antenna modules and antenna solutions.
OriginGPS modules introduce unparalleled sensitivity and noise immunity by incorporating Noise Free Zone system (NFZ™) proprietary technology for faster position fix and navigation stability even under challenging satellite signal conditions.
Founded in 2006, OriginGPS is specializing in development of unique technologies that miniaturize RF modules, thereby addressing the market need for smaller wireless solutions.

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1.3 Nano-spider Architecture

